6CCSACCM
Cooperation, Coordination and Multi-Agent Systems

Week 4

Cooperative games (part I)

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Binding agreements

- Cooperation does not occur in prisoner's dilemma, because players cannot make binding agreements.
- But what if binding agreements are possible?
- This is the class of scenarios studied by *cooperative game theory*.

Cooperative games

- **Cooperative** (or, **coalitional**) **games** model scenarios, where
  - agents can benefit by cooperating;
  - binding agreements are possible.
- In cooperative games, actions are taken by *coalitions* of agents.

TU vs. NTU games

- **Transferable utility** (TU) games: payoffs are given to the group and then divided among its members.
- **Non-transferable utility** (NTU) games: group actions result in payoffs to individual group members.

Phases of coalitional action

- Agents form coalitions (teams).
- Each coalition chooses its action.
- NTU games: the choice of coalitional actions (by all coalitions) determines each player's payoff.
- TU games: the choice of coalitional actions (by all coalitions) determines the payoff of each coalition.
  - The members of the coalition then need to divide this joint payoff.

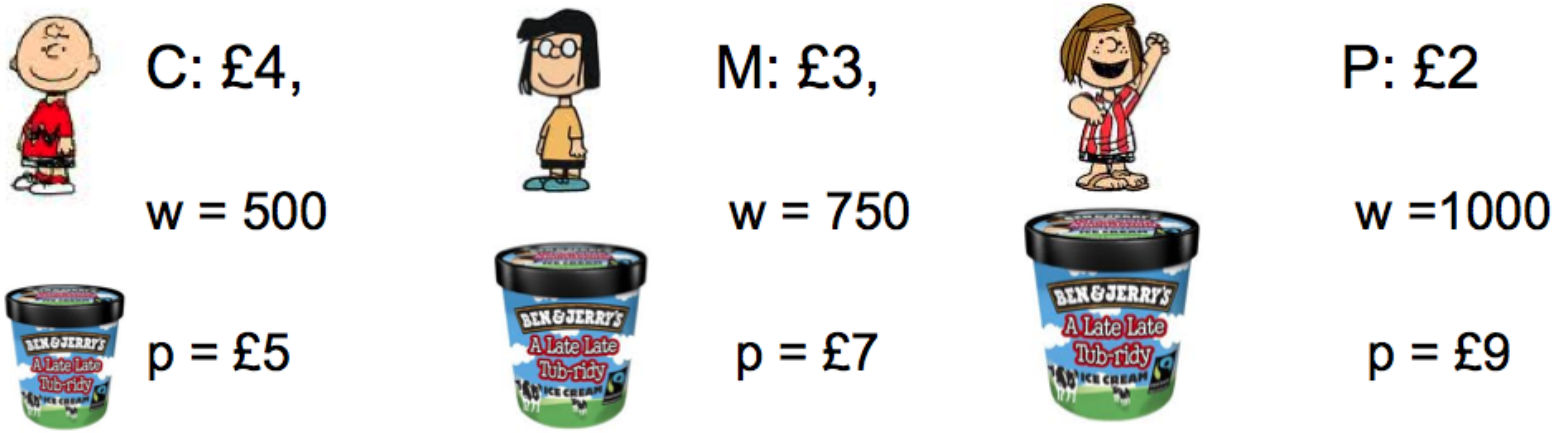
NTU game example: academic collaboration

- n researchers working at n different universities can form groups to collectively write papers.
- The composition of a group determines the quality of the paper they produce.
- Each author receives a payoff from their own university:
  - promotion;
  - bonus;
  - teaching load reduction;
  - a parking spot.
- Payoffs are *non-transferable*.

TU game example: farmers' cooperation

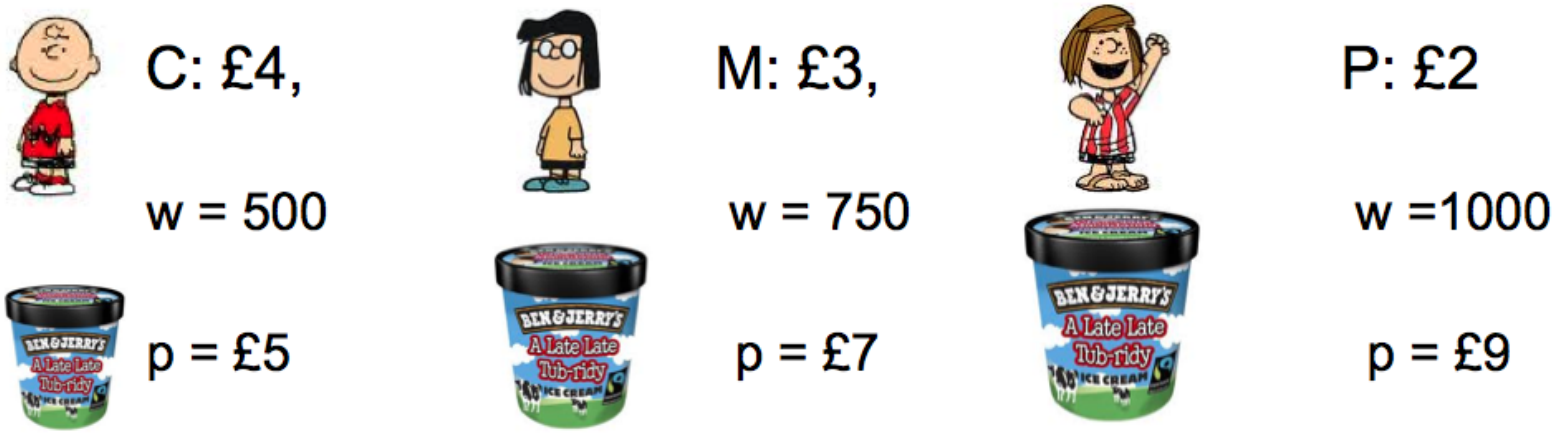
- n farmers can cooperate to grow fruit.
- A group of farmers can grow apples or oranges.
  - A group of size k produces $f(k)$ tons of apples and $g(k)$ tons of oranges.
- Fruit is sold in the market.
  - If there are x tons of apples and y tons of oranges on the market, the market prices for apples and oranges are $p(x)$ and $q(y)$, respectively.
- The profit of a group depends on the quantity and type of fruit it grows, and the market price.

TU game example: buying ice-cream



- n children, each has some amount of money.
- Three types of ice-cream tubs are for sale.
- Children have utility for ice-cream, and do not care about money.

TU game example: buying ice-cream



- The payoff of each group: the maximum quantity of ice-cream the members of the group can buy by pooling their money.
- The ice-cream can be shared arbitrarily among the group members.

Characteristic function vs. partition function

- In general TU games, the payoff obtained by a coalition depends on the actions chosen by other coalitions.
  - These games are called **partition function games** (PFG).
- **Characteristic function games** (CFG): the payoff of each coalition only depends on the action of that coalition.
  - In such games, each coalition can be identified with the profit it obtains by choosing its best action.
- Ice-cream game is a CFG. Farmers' cooperation is a PFG.

How is a cooperative game played?

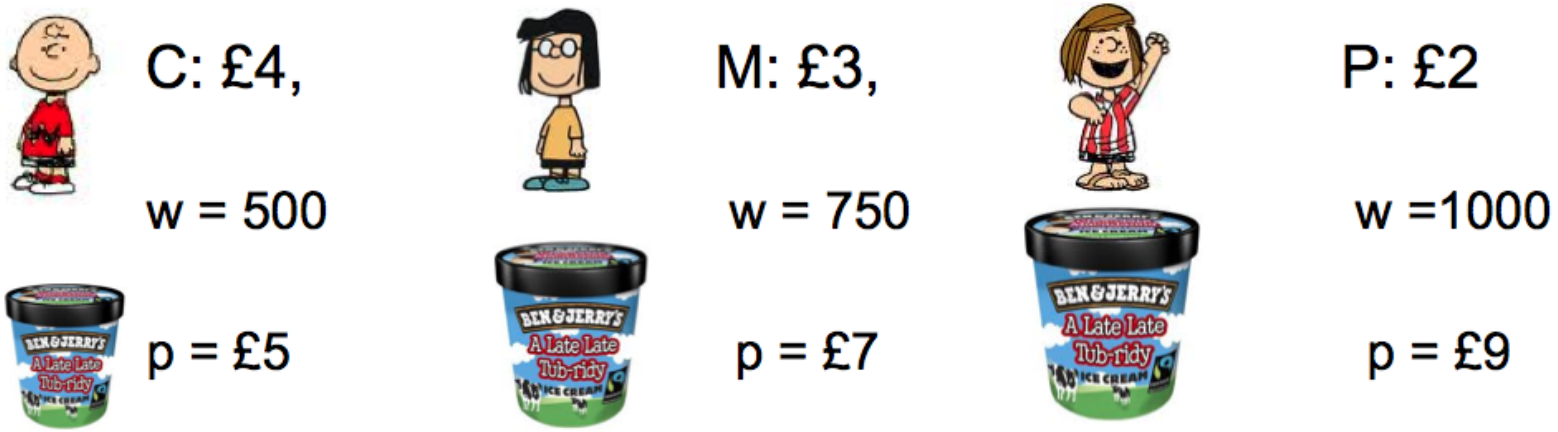
- Even though agents work together they are still *selfish*.
- The partition into coalitions and payoff distribution should be such that no player (or group of players) has an incentive to deviate.
- We may also want to ensure that the outcome is fair: the payoff is proportional to the agent's contribution.
- We will now see how to formalise these ideas.

TU games formalised

A **transferable utility game** is a pair (N, v), where:

- $N = \{1, \ldots, n\}$ is a set of players.
- $v : 2^N \to \mathbb{R}$ is a characteristic function.
  - For each subset of players C, $v(C)$ is the amount that the members of C can earn by working together.
- Usually it is assumed that v is
  - *normalised*: $v(\emptyset) = 0$;
  - *non-negative*: $v(C) \geq 0$ for any $C \subseteq N$;
  - *monotone*: $v(C) \leq v(D)$ for any C, D such that $C \subseteq D$.
- A *coalition* is any subset of N.
  - N itself is called the *grand coalition*.

Example: ice-cream game



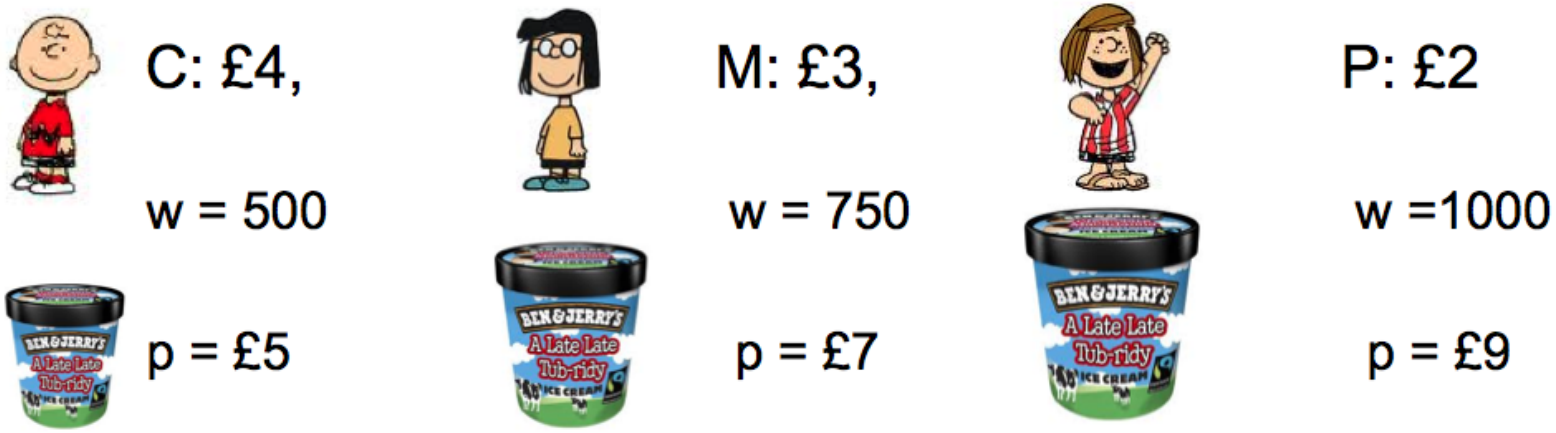
- $v(\emptyset) = v(\{C\}) = v(\{M\}) = v(\{P\}) = 0$
- $v(\{C, P\}) = v(\{M, P\}) = 500$, $v(\{C, M\}) = 750$
- $v(\{C, M, P\}) = 1000$

TU games: outcome

An **outcome** of a TU game (N, v) is a pair (CS, x), where:

- $CS = (C_1, \ldots, C_k)$ is a *coalition structure* – a partition of N into coalitions:
  - $\cup_{i=1}^{k} C_i = N$ and $C_i \cap C_j = \emptyset$ for $j \neq j$;
- $x = (x_1, \ldots, x_n)$ is a *payoff vector*, which distributes the value of each coalition in CS:
  - $x_i \geq 0$ for all $i \in N$;
  - $\sum_{i \in C} x_i = v(C)$ for each C in CS.

Example: ice-cream game



- $CS = (\{C\}, \{M\}, \{P\})$; $x_C = 0, x_M = 0, x_P = 0$
- $CS = (\{C\}, \{M, P\})$; $x_C = 0, x_M = 150, x_P = 350$
- $CS = (\{C, M, P\})$; $x_C = 334, x_M = 333, x_P = 333$
- $CS = (\{C, M, P\})$; $x_C = 0, x_M = 800, x_P = 200$

TU games: outcome

- An outcome (CS, x) is called an **imputation** if it satisfies individual rationality:
  - $x_i \geq v(\{i\})$ for all $i \in N$.
- Notation: we will denote $\sum_{i \in C} x_i$ by $x(C)$.

Example: weighted voting games

- n players with weights $w_1, \ldots, w_n$, quota q;
- $w(C) = \sum_{i \in C} w_i$;
- $v(C) = 1$ (C wins) if $w(C) \geq q$; $v(C) = 0$ (C loses) otherwise.

Weighted voting games

United Kingdom, 2010:

- 650 seats, $q = 326$
  - Conseratives (C): 306
  - Labour (L): 258
  - Liberal Democrats (LD): 57
  - Scottish National Party (SNP): 6
  - Democratic Unionist Party (DUP): 8
  - 6 other parties (O), with a total of 15 seats
- $N = \{C, L, LD, SNP, DUP, O\}$
- $v(\{C, L\}) = v(\{C, LD\}) = v(\{C, SNP, DUP, O\}) = 1$
- $v(\{L, LD\}) = 0$, $v(\{L, LD, SNP, DUP\}) = 1$

Example: network flow games

- network $G = (V, E)$, with source s and sink t, and edge capacities $c_1, \ldots, c_{|E|}$;
- players are edges;
- $v(C)$ is the maximum $s - t$ flow that C can carry.

Example: graph induced games

- players are vertices of a weighted graph
- value of a coalition = total weight of internal edges
- model for social networks
- weights need not be positive!
- *example outcome*: grand coalition forms, each player receives half of the weight of adjacent edges

Special classes: simple games

A game $G = (N, v)$ is **simple** if:

- value of each C is 0 or 1: $v(C) \in \{0, 1\}$ for any $C \subseteq N$
- v is *monotone*: if $S \subseteq T$ then $v(S) \leq v(T)$.
  - A coalition C in a simple game is said to be *winning* if $v(C) = 1$ and *losing* if $v(C) = 0$.
  - *Example:* weighted voting games with non-negative weights (but not general weighted voting games).

<u>Convention:</u> in simple games, unless specified otherwise, we assume that players form the *grand coalition*.

Simple games: player types

- In a simple game, a player $i \in N$ is a **null** player if: $v(C) = v(C \cup \{i\})$ for any $C \subseteq N \setminus \{i\}$.
- In a simple game, a player $i \in N$ is a **veto** player if: $v(C) = 0$ for any $C \subseteq N \setminus \{i\}$.
  - Equivalently, by monotonicity, $v(N \setminus \{i\}) = 0$.

Weighted voting games

United Kingdom, 2010:

- 650 seats, $q = 326$
  - Conseratives (C): 306
  - Labour (L): 258
  - Liberal Democrats (LD): 57
  - Scottish National Party (SNP): 6
  - Democratic Unionist Party (DUP): 8
  - 6 other parties (O), with a total of 15 seats
- no player is a null player
- no player is a veto player

Weighted voting games

United Kingdom, 2017:

- 650 seats, $q = 326$
  - Conseratives (C): 317
  - Labour (L): 262
  - Liberal Democrats (LD): 12
  - Scottish National Party (SNP): 35
  - Democratic Unionist Party (DUP): 10
  - 6 other parties + speaker (O), with a total of 14 seats
- $N = \{C, L, LD, SNP, DUP, O\}$
- $V(C) = 0$, $v(\{C, L\}) = v(\{C, SNP\}) = \ldots = v(\{C, DUP\}) = 1$
- $v(\{L, LD, SNP, DUP, O\}) = 1$

Weighted voting games

United Kingdom, 2019:

- 650 seats, $q = 326$
  - Conseratives (C): 365
  - Labour (L): 202
  - Liberal Democrats (LD): 11
  - Scottish National Party (SNP): 47
  - Democratic Unionist Party (DUP): 8
  - 6 other parties + speaker(O), with a total of 17 seats
- $N = \{C, L, LD, SNP, DUP, O\}$
- C is a veto player, all others are null players

Special classes: superadditive games

A game $G = (N, v)$ is **superadditive** if $v(S \cup T) \geq v(S) + v(T)$ for any disjoint $S, T \subseteq N$.

- Two coalitions can always merge without losing money; hence, we can assume that players form the grand coalition.
- *Example:* Network flow games.

Superadditive games

- Convention: in superadditive games, we identify outcomes with payoff vectors for the grand coalition.
  - An outcome is a vector $x = (x_1, \ldots, x_n)$ with $\sum_{i \in N} x_i = v(N)$.
- Caution: some literature defines outcomes in this way even if the game is not superadditive.

Special classes: convex games

A function $f : 2^N \to \mathbb{R}$ is called **supermodular** if $f(\emptyset) = 0$ and $f(A \cup B) + f(A \cap B) \geq f(A) + f(B)$ for any $A, B \subseteq N$ (not necessarily disjoint).

- Any supermodular function is superadditive, but the converse is not true.

A TU game $G = (N, v)$ is **convex** if its characteristic function is supermodular.

Special classes: convex games

Proposition

If f is supermodular, $T \subset S$ and $i \notin S$ then
$f(T \cup \{i\}) - f(T) \leq f(S \cup \{i\}) - f(S)$.

- A player is more useful when he joins a bigger coalition.

Proof.

Let $A = S$ and $B = T \cup \{i\}$. Then, $A \cup B = S \cup \{i\}$ and $A \cap B = T$. $\square$

Ice-cream games: properties

Are ice-cream games superadditive?

- Yes.

Are ice-cream games convex?

- No.

Graph-induced games: properties

Are graph-induced games superadditive?

- If and only if the weight of each edge is non-negative.

Are graph-induced games convex?

- If and only if the weight of each edge is non-negative.

Weighted voting games: properties

Are weighted voting games superadditive?

- No (in general): $N = \{1, 2\}$; $w_1 = w_2 = 1$; $q = 1$.
- Yes, if $q > \frac{w(N)}{2}$

Are weighted voting games convex?

- Not necessarily, even if $q > \frac{w(N)}{2}$.
- *Example:* $N = \{1, 2, 3\}$; $w_1 = w_2 = w_3 = 1$; $q = 2$;

$$v(\{1, 2\}) + v(\{2, 3\}) = 2 > 1 = v(\{1, 2, 3\}) + v(\{2\}).$$

What is a good outcome?

C: £4,

w = 500

p = £5

M: £3,

w = 750

p = £7

P: £2

w =1000

p = £9

- $v(\emptyset) = v(\{C\}) = v(\{M\}) = v(\{P\}) = 0$
- $v(\{C, P\}) = v(\{M, P\}) = 500$, $v(\{C, M\}) = 750$
- $v(\{C, M, P\}) = 1000$

What is a good outcome?

C: £4,

w = 500

p = £5

M: £3,

w = 750

p = £7

P: £2

w =1000

p = £9

- This is a superadditive game.
  - Outcomes are payoff vectors.
- How should the players share the ice-cream?
  - If they share as $(350, 350, 300)$, C and M can get more ice-cream by buying a 750g tub on their own, and splitting it equally.
  - $\Rightarrow$ The outcome (350, 350, 300) is not stable!

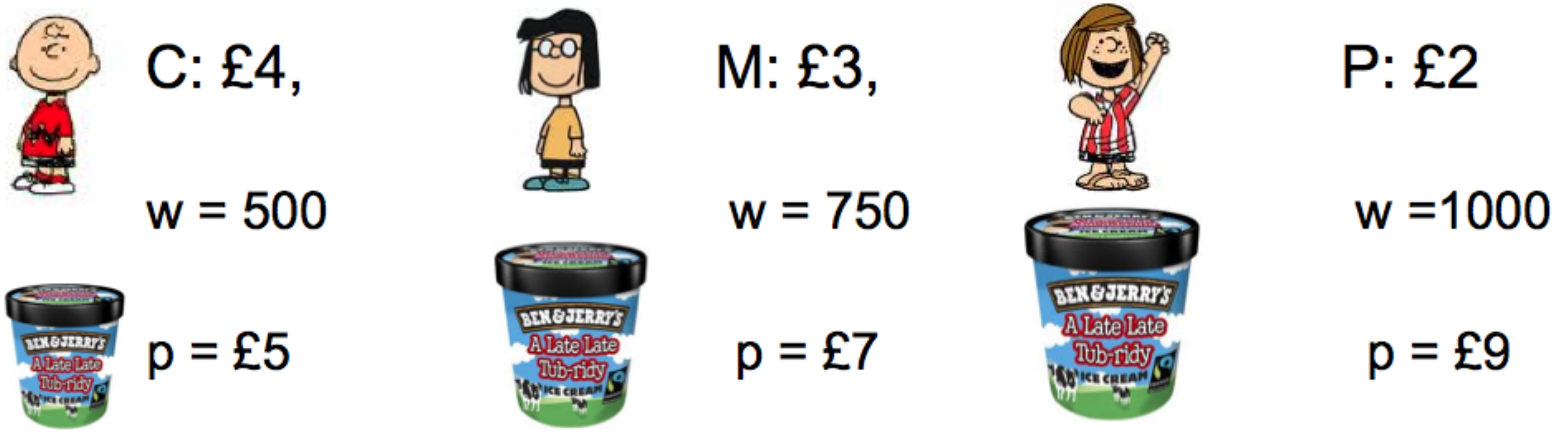
Core and core-related concepts

Definition

The core $\mathcal{C}(G)$ of a characteristic function game $G = (N, v)$ is the set of all outcomes $(CS, \mathbf{x})$ such that $x(C) \geq v(C)$ for every $C \subseteq N$.

- Each coalition earns at least as much as it can make on its own.
- Note that G is not assumed to be superadditive.

Example: ice-cream game



- $v(\emptyset) = v(\{C\}) = v(\{M\}) = v(\{P\}) = 0$
- $v(\{C, P\}) = v(\{M, P\}) = 500$, $v(\{C, M\}) = 750$
- $v(\{C, M, P\}) = 1000$

Example: ice-cream game

For any outcome $(CS, (x_C, x_M, x_P))$ in the core, it has to be $CS = \{N\}$ (since the game is super-additive).

- Otherwise, we would have

$$x(N) = \sum_{C \in \mathcal{CS}} v(C) < v(N)$$

violating the core constraint for N.

Example: ice-cream game

Furthermore, the values x_C, x_M, x_P have to satisfy the following constraints:

$$
\begin{aligned}
x_C \geq 0, \quad x_M \geq 0, \quad x_P &\geq 0 \\
x_C + x_M + x_P &= 1000 \\
x_C + x_M &\geq 750 \\
x_C + x_P &\geq 500 \\
x_M + x_P &\geq 500
\end{aligned}
$$

The system has several solutions. For example:

$$
\begin{aligned}
&(400, 400, 200) \\
&(500, 400, 100) \\
&(500, 500, 0)
\end{aligned}
$$

Core stability implies efficiency

- The outcomes in the core are stable in the sense that no coalition has incentives to deviate as it cannot do better on its own.
  - Therefore, such outcomes are more likely to arise when a coalitional game is played.
- Stability implies efficiency:
  - any outcome in the core maximises the *social welfare* – the total payoff to all the players.

Core stability implies efficiency

Theorem

If an outcome $(CS, \mathbf{x})$ is in the core of a characteristic function game $G = (N, v)$ then $v(CS) \geq v(CS')$ for every coalition structure $CS' \in \mathcal{CS}_N$, where $\mathcal{CS}_N$ denotes the space of all coalition structures over N.

Core stability implies efficiency

Proof.

- Assume on the contrary that $v(CS) < v(CS')$ for some coalition structure $CS' \in \mathcal{CS}_N$.
- Then,

$$\sum_{C' \in CS'} x(C') = \sum_{i \in N} x_i = v(CS) < v(CS') = \sum_{C' \in CS'} v(C')$$

- On the other hand, since $(CS, \mathbf{x})$ is in the core, we have $x(C') \geq v(C')$ for any $C' \in CS'$, and therefore

$$\sum_{C' \in CS'} x(C') \geq \sum_{C' \in CS'} v(C')$$

– a contradiction.

$\square$

Example: empty core

Recall the majority game with 3 players: WVG$[2; 1, 1, 1]$

- $v(C) = 1$ for $|C| \geq 2$;
- $v(C) = 0$ for $|C| \leq 1$.

Consider an outcome $(CS, \mathbf{x})$:

- if $CS = \{\{1\}, \{2\}, \{3\}\}$, the grand coalition can deviate;
- if $CS = \{\{1, 2\}, \{3\}\}$, either 1 or 2 gets payoff less than 1, so can deviate with player 3;
- same argument for $CS = \{\{1, 3\}, \{3\}\}$ and $CS = \{\{2, 3\}, \{1\}\}$;
- suppose $CS = \{\{1, 2, 3\}\}$: then, $x_i > 0$ for some i, so $x(N \setminus \{i\}) < 1$, yet $v(N \setminus \{i\}) = 1$.

Example: ice-cream game

C: £4,

w = 500

p = £5

M: £3,

w = 750

p = £7

P: £2

w =1000

p = £9

- $(500, 500, 0)$ is a core outcome.
- But is it fair?

Shapley value

- One of the solution concepts that capture the notion of fairness is the **Shapley value**.
- The Shapley value is based on the intuition that the payment that each agent receives should be *proportional to its contribution*.

Shapley value: attempts

- A naive implementation of this idea would be to pay each agent according to how much he increases the value of the coalition of all other players when it joins in – i.e., set the payoff of the player i to $v(N) - v(N \setminus \{i\})$.
- However, under this payoff scheme the total payoff assigned to the agents may differ from the value of the grand coalition: e.g., in the 3-player majority game, each agent's payoff under this scheme would be 0, whereas the value of the grand coalition is 1.

Shapley value: attempts

- We can fix an ordering of the agents and pay each agent according to how much it contributes to the coalition formed by its predecessors in this ordering:
  - that is, agent 1 receives $v(\{1\})$, agent 2 receives $v(\{1,2\}) - v(\{1\})$, etc.
- It is easy to see that this payoff scheme distributes the value of the grand coalition among the agents.

Shapley value: attempts

- However, it suffers from another problem: two agents that play symmetric roles in the game may receive very different payoffs.
  - In the 3-player majority game, under the order-based payoff scheme one agent (the one that happens to be the second in the ordering) will receive 1, whereas two other agents receive 0, even though all three agents are clearly identical in terms of their contribution.
- In other words, the payoffs in this scheme strongly depend on the selected ordering of the agents.

Shapley value

This dependence can be eliminated by *averaging* over all possible orderings (or, **permutations**) of the players.

- Let Π_N denote the set of all permutations of N.
- Given a permutation $\pi \in \Pi_N$, let $S_\pi(i)$ be the set of all predecessors of i in π:

$$S_\pi(i) = \{j \in N | \pi(j) < \pi(i)\}$$

Shapley value

For example, if $N = \{1, 2, 3\}$ then

$$\Pi_N = \{(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)\}$$

For $\pi = (3, 2, 1)$ we have:

- $S_\pi(3) = \emptyset$,
- $S_\pi(2) = \{3\}$, and
- $S_\pi(1) = \{2, 3\}$.

Shapley value

The **marginal contribution** of an agent i with respect to a permutation π in a game $G = (N, v)$ is given by

$$\Delta_\pi^G(i) = v\left(S_\pi(i) \cup \{i\}\right) - v\left(S_\pi(i)\right)$$

Definition

Given a characteristic function game $G = (N, v)$ with $|N| = n$, the **Shapley value** of a player $i \in N$ is given by

$$\phi_i(G) = \frac{1}{n!} \sum_{\pi \in \Pi_N} \Delta_\pi^G(i)$$

Shapley value: probabilistic interpretation

ϕ_i is i's average marginal contribution to the coalition of its predecessors, over all permutations

- Suppose that we choose a permutation of players uniformly at random, among all possible permutations of N. Then, ϕ_i is the *expected marginal contribution* of player i to the coalition of its predecessors.

Example: ice-cream game

- Let's calculate the Shapley value for Pattie.
- There are six permutations of the players:

$$\pi_1 = CMP \quad \pi_2 = MCP \quad \pi_3 = CPM$$
$$\pi_4 = MPC \quad \pi_5 = PCM \quad \pi_6 = PMC$$

- We have:

$$\begin{aligned}
\Delta^G_{\pi_1}(i) = \Delta^G_{\pi_2}(i) &= v(N) - v(\{C, M\}) = 250 \\
\Delta^G_{\pi_3}(i) &= v(\{C, P\}) - v(\{C\}) = 500 \\
\Delta^G_{\pi_4}(i) &= v(\{M, P\}) - v(\{M\}) = 500 \\
\Delta^G_{\pi_5}(i) = \Delta^G_{\pi_6}(i) &= v(\{P\}) - v(\emptyset) = 0
\end{aligned}$$

Hence, $\phi_P(G) = \frac{250+250+500+500}{6}$.

Shapley value: properties

The Shapley value is **efficient**, i.e., it distributes the value of the grand coalition among all agents.

- For any characteristic function game $G = (N, v)$,

$$\sum_{i \in N} \phi_i(G) = v(N)$$

Shapley value: properties

Another useful property of the Shapley value is that it does not allocate any payoffs to players who do not contribute to any coalition.

- Given a characteristic function game $G = (N, v)$, a player i is said to be a **null player** if $v(C \cup \{i\}) = v(C)$ for any $C \subseteq N$.
- If $i \in N$ is a null player in G, then $\phi_i(G) = 0$.

Shapley value: properties

Another important observation is that if two players contribute equally to each coalition,then their Shapley values are equal.

- Given a characteristic function game $G = (N, v)$, we say that players i and j are **symmetric** in G if

$$v(C \cup \{i\}) = v(C \cup \{j\})$$

  for any $C \subseteq N \setminus \{i, j\}$.
- If players i and j are symmetric in G, then $\phi_i(G) = \phi_j(G)$.

Shapley value: properties

Consider a group of players N that is involved in two coalitional games G^1 and G^2:

- $G^1 = (N, v^1)$;
- $G^2 = (N, v^2)$.

The **sum** of G^1 and G^2 is a coalitional game $G^+ = G^1 + G^2$ given by $G^+ = (N, v^+)$, where for any coalition $C \subseteq N$,

$$v^+(C) = v^1(C) + v^2(C)$$

For any player i, we have

$$\phi_i(G^+) = \phi_i(G^1) + \phi_i(G^2)$$

Shapley value: properties

To summarise, we have shown that the Shapley value possesses four desirable properties:

1. **Efficiency**: $\sum_{i \in N} \phi_i(G) = v(N)$;
2. **Dummy player**: if $i \in N$ is a dummy, then $\phi_i(G) = 0$;
3. **Symmetry**: if i and j are symmetric, then $\phi_i(G) = \phi_j(G)$;
4. **Additivity**: $\phi_i(G^1 + G^2) = \phi_i(G^1) + \phi_i(G^2)$.

- Shapley value is the *only* payoff division scheme that has these four properties.
- These properties can also be used to simplify the computation of the Shapley value.

Example

Consider a game $G = (N, v)$ with $|N| = n$ players, where

- $v(N) = 1$;
- $v(C) = 0$ for any $C \subsetneq N$.

We can compute the Shapley value of a player $i \in N$ by observing that $\Delta_\pi^G(i) = 1$ if i appears in the last position in π and $\Delta_\pi^G(i) = 0$ otherwise. Since there're exactly $(n-1)!$ permutations with player i in the last position, we have

$$\phi_i(G) = \frac{(n-1)!}{n!} = \frac{1}{n}$$

Alternatively, we can simply observe that all players are symmetric and hence $\phi_1(G) = \ldots = \phi_n(G)$. Now, by the efficiency property, for all $i \in N$ this implies

$$\phi_i(G) = \frac{v(N)}{n} = \frac{1}{n}$$

Banzhaf index

Another solution concept which is motivated by fairness considerations is the *Banzhaf index*.

- Just like the Shapley value, it measures agents' expected marginal contributions.
- However, instead of averaging over all permutations of players, it averages over all coalitions in the game.

Definition

Given a characteristic function game $G = (N, v)$ with $|N| = n$, the **Banzhaf index** of a player $i \in N$ is given by

$$\beta_i(G) = \frac{1}{2^{n-1}} \sum_{C \subseteq N \setminus \{i\}} [v(C \cup \{i\}) - v(C)]$$

Banzhaf index

- Banzhaf index satisfies properties (2)-(4).
- However, it does not satisfies property (1) – efficiency.

Example

Consider again the game $G = (N, v)$ with $|N| = n$ players, where

- $v(N) = 1$;
- $v(C) = 0$ for any $C \subsetneq N$.

Fix a player i. We have:

- $v(C \cup \{i\}) - v(C) = 1$ if $C = N \setminus \{i\}$, and
- $v(C \cup \{i\}) - v(C) = 0$ for any proper subset C of $N \setminus \{i\}$.

Therefore, $\beta_i(G) = \frac{1}{2^{n-1}}$ for each $i \in N$, and hence for $n > 2$,

$$\sum_{i \in N} \beta_i(G) < v(N)$$

Normalised Banzhaf index

To ensure efficiency, we can normalise the Banzhaf index as follows:

$$\nu_i(G) = \frac{\beta_i(G)}{\sum_{i \in N} \beta_i(G)}$$

However, this version of the Banzhaf index loses additivity.

Shapley value and Banzhaf index in simple games

In simple games, Shapley value and Banzhaf index have a particularly attractive interpretation:

- they measure the **power** of a player – the probability that he can influence the outcome of the game.
- The Shapley value is often referred to as the *Shapley-Shubik power index* in this case.

THANK YOU!